Spacer composition dominates apparent regulatory grammar signals detected by DNA sequence models

Motivation

Whether the spatial arrangement of transcription factor (TF) binding sites encodes functional information—a "regulatory grammar"—or whether enhancers operate as "billboards" where only TF identity matters remains debated [1,2]. DNA sequence models enable systematic *in silico* perturbation of enhancer architecture, but standard grammar sensitivity metrics may confound motif arrangement with spacer DNA composition. We introduce a **spacer-factored grammar sensitivity framework (SF-GSI)** to separate these effects across four model architectures, five MPRA datasets, and four organisms.

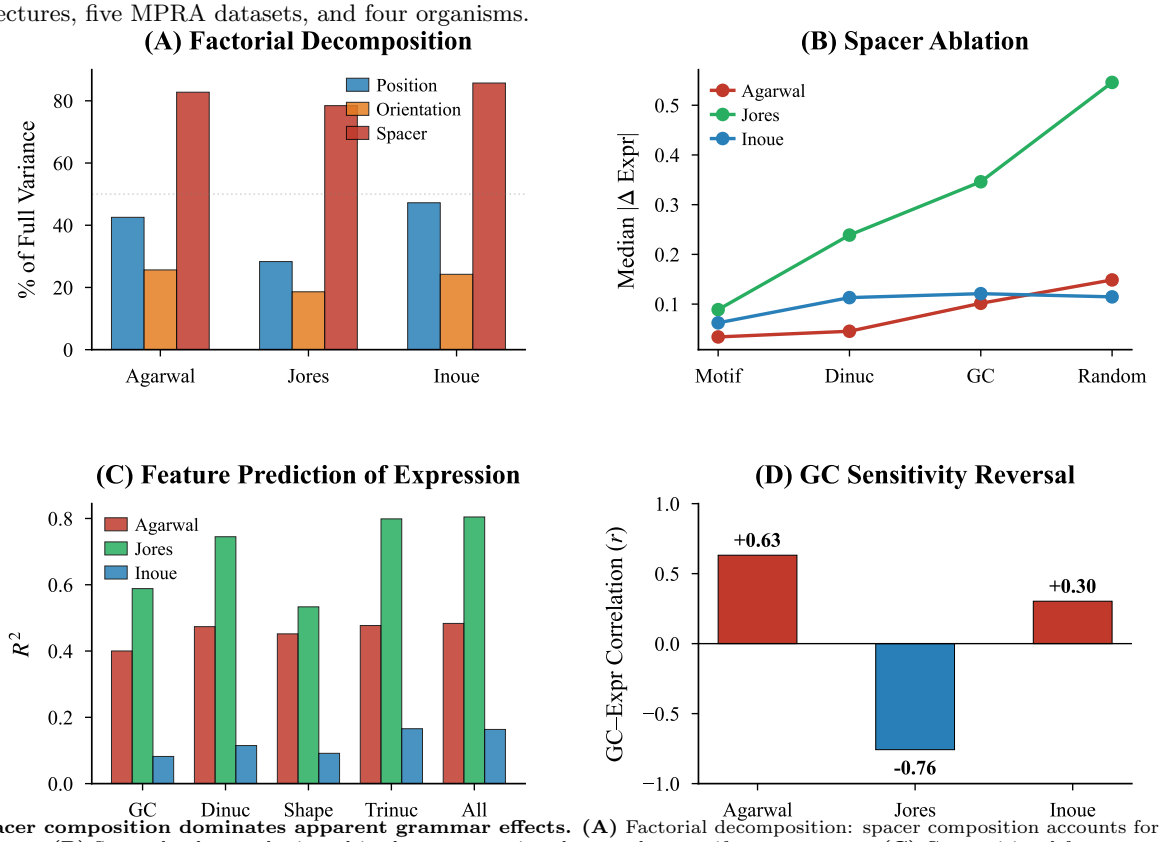

Figure 1: Spacer composition dominates apparent grammar effects. (A) Factorial decomposition: spacer composition accounts for the majority of expression variance. **(B)** Spacer-level perturbations drive larger expression changes than motif rearrangement. **(C)** Compositional features predict 40–80% of model R^2. **(D)** GC–expression correlation reverses across species ($r = +0.66$ human vs. $r = -0.73$ plant).

Methods

Grammar Sensitivity Index (GSI). TF motifs are identified via FIMO against JASPAR 2024 [3]; vocabulary-preserving shuffles permute motif order while holding motif identity constant. GSI $= \sigma(\hat{y})/|\mu(\hat{y})|$. **Spacer-Factored GSI (SF-GSI).** A factorial design isolates: (i) motif position/orientation (true grammar), (ii) spacer composition (confound), (iii) flanking-context effects via matched-spacer shuffles and regression residuals. **Rule Extraction.** Pairwise rules via spacing scans (2–50 bp), four orientation configurations, and helical phasing at ∼10.5 bp. **Variance Decomposition.** Hierarchical ANOVA: vocabulary → +grammar → +higher-order → full model. **Models & Data.** DNABERT-2, Nucleotide Transformer v2-500M, HyenaDNA, Enformer, and MPRA-trained PARM; five MPRA datasets across human, yeast, and plant with 7,650 model–enhancer measurements in the main census plus controlled synthetic positive controls [4,5].

Results

The spacer confound. Across datasets, spacer DNA composition accounts for 78–86% of measured GSI variance (Fig. 1A–B). Simple compositional statistics explain 40–80% of model predictions (Fig. 1C), and the GC–expression relationship reverses between species (Fig. 1D).

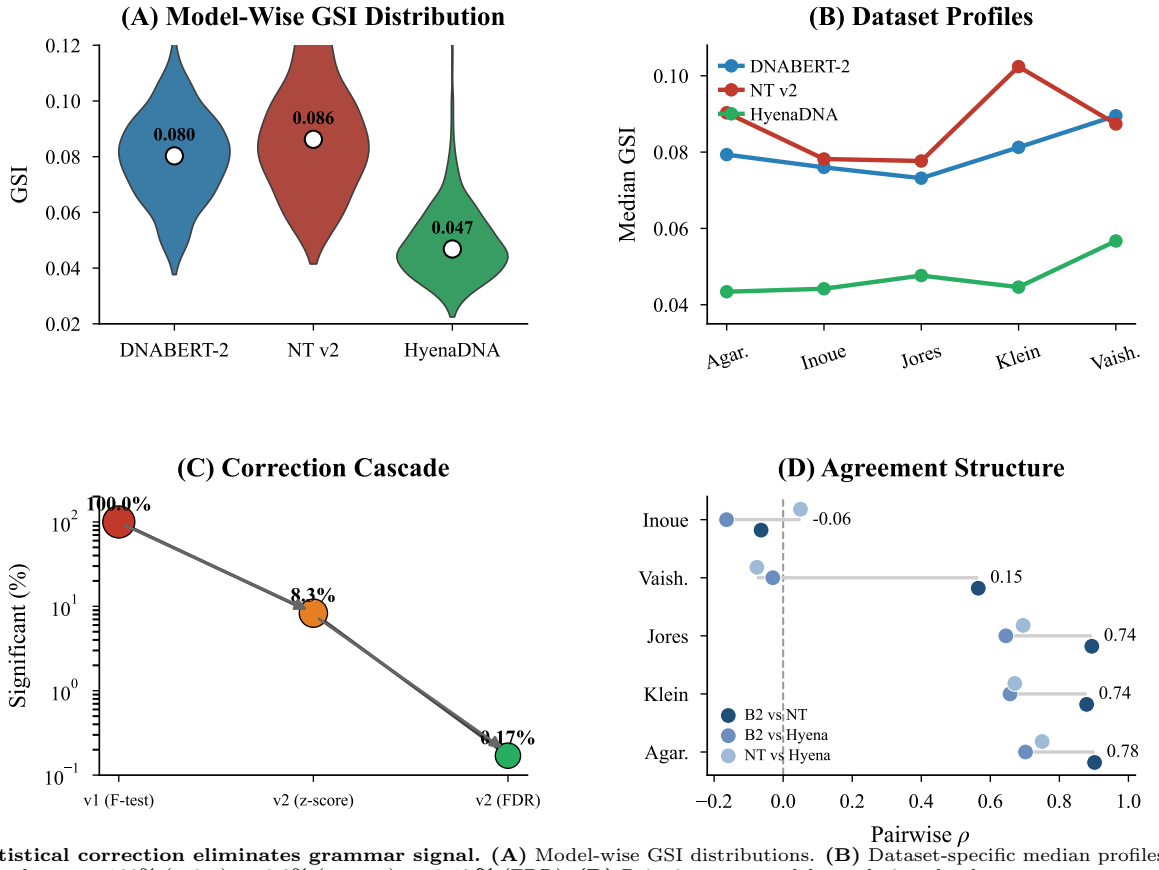

Figure 2: Statistical correction eliminates grammar signal. (A) Model-wise GSI distributions. **(B)** Dataset-specific median profiles across models. **(C)** Significant enhancers: 100% (naive) → 8.3% (z-score) → **0.17%** (FDR). **(D)** Pairwise cross-model correlations by dataset.

Correction cascade. In the 7,650-measurement census, naive thresholding marks all enhancers as grammar-sensitive, but null-normalized testing reduces this to 8.3%, and FDR correction to **0.17%** (13/7,550)—a three-order-of-magnitude drop consistent across architectures (Fig. 2C–D).

Positive controls. When spacer sequence is held constant in controlled synthetic constructs, orientation effects remain detectable across models ($p \leq 7.2 \times 10^{-15}$; two models reach numerical underflow), confirming that the framework can recover genuine grammar effects when the confound is removed.

Transfer & variance decomposition. Grammar rules from one organism predict near-zero effects in others. In ANOVA decompositions, motif vocabulary explains 8–22% of variance, whereas arrangement contributes only 0–1.6%. Learned embeddings outperform explicit grammar features by 16–42 percentage points of R^2, indicating that current motif-centric grammar summaries miss most predictive signal.

Conclusion

Across these datasets and models, most apparent regulatory grammar signal arises from spacer composition rather than motif arrangement. Genuine grammar effects are still detectable in spacer-controlled positive controls, but standard vocabulary-preserving shuffles substantially overestimate their prevalence. SF-GSI provides a practical route to separate grammatical effects from compositional confounds in regulatory genomics and offers a stronger benchmark for evaluating sequence models as mechanistic predictors. Code: `https://github.com/bryanc5864/GRAMLANG`.

References: [1] Arnosti & Kulkarni, *J. Cell. Biochem.* 94:890, 2005. [2] Long et al., *Cell* 167:1170, 2016. [3] Castro-Mondragon et al., *Nucleic Acids Res.* 52:D142, 2024. [4] Agarwal et al., *Science* 380:eadh0559, 2023. [5] Jores et al., *Nat. Plants* 7:842, 2021.